

Two-phase simplex

Simplex requires a BFS to start. What if there is no obvious BFS? Create another LP with an obvious BFS, and use simplex to determine if the original LP has a BFS.

Phase I: Either find a BFS, or show that the LP is infeasible.

Form an auxiliary LP that has an obvious BFS.

① If RHS is negative, multiply the row by -1.

$$\begin{pmatrix} 3 & -1 & 7 & 1 \\ 1 & 0 & -2 & 0 \\ 5 & 2 & 5 & 3 \end{pmatrix} x = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} 3 & -1 & 7 & 1 \\ -1 & 0 & 2 & 0 \\ 5 & 2 & 5 & 3 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

② Add auxiliary variables to form an identity.

$$\rightarrow \begin{pmatrix} 3 & -1 & 7 & 1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 1 & 0 \\ 5 & 2 & 5 & 3 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

We now have an obvious BFS.

Basis $\{5, 6, 7\}$, BFS $(0, 0, 0, 0, 3, 1, 2)^T$.

Observation: If we can find a BFS where all auxiliary variables are 0's, then we have found a BFS for the original LP (just remove the aux vars). Converse also true.

③ Add objective function minimizing the aux vars.

$$\min x_5 + x_6 + x_7. \rightarrow \max -x_5 - x_6 - x_7.$$

Auxiliary LP for the example:

$$\begin{aligned} \max & (0 \ 0 \ 0 \ 0 \ -1 \ -1 \ -1) x \\ \text{s.t.} & \begin{pmatrix} 3 & -1 & 7 & 1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 1 & 0 \\ 5 & 2 & 5 & 3 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

[Note: Not in canonical form.]

We then solve the auxiliary LP using Simplex.

- If the optimal value is 0, then we found a solution where all aux vars are 0's.

This gives a feasible solution to the LP. Phase II is regular simplex starting with our feasible solution.

- If the optimal value is negative, then it is impossible to force all aux vars to 0's. So the original LP is infeasible.

Certificate of infeasibility

If the optimal value of the aux LP is negative, then $y = A_B^{-T} c_B$ is a certificate of infeasibility for the LP after we multiplied rows by -1 .

Example: If $\{2, 5, 6\}$ is the final basis with negative optimal value, then

$$y = A_B^{-T} c_B = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}^{-T} \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$$

$$\begin{array}{ll} \max & (0 \quad 0 \quad 0 \quad 0 \quad -1 \quad -1) x \\ \text{s.t.} & \begin{pmatrix} 3 & -1 & 7 & 1 & 1 & 0 \\ -1 & 0 & 2 & 0 & 0 & 0 \\ 5 & 2 & 5 & 3 & 0 & 0 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \\ & x \geq 0 \end{array}$$

is a certificate of infeasibility for the LP $\begin{pmatrix} 3 & -1 & 7 & 1 \\ -1 & 0 & 2 & 0 \\ 5 & 2 & 5 & 3 \end{pmatrix} x \leq \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$.

If we want the certificate for the original LP $\begin{pmatrix} 3 & -1 & 7 & 1 \\ -1 & 0 & 2 & 0 \\ 5 & 2 & 5 & 3 \end{pmatrix} x \leq \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$, then

we multiply the second entry of y by -1 .